VaultSign AI

Project Antigravity: Zero-Trust Contract Negotiation Platform

1. EXECUTIVE OVERVIEW

Independent contractors and freelancers frequently face complex, jargon-heavy client contracts containing predatory clauses. However, they cannot afford expensive legal counsel for every gig, and critically, they cannot upload these confidential documents to cloud-based AI tools (like ChatGPT) without directly violating the strict Non-Disclosure Agreements (NDAs) embedded within the contracts themselves.

VaultSign AI solves this NDA paradox. By leveraging local inference and open-weight language models, VaultSign acts as a zero-trust legal co-pilot. It ingests complex documents locally, highlights red flags, and generates professional counter-offer emails—all without a single byte of confidential data ever leaving the freelancer's machine.

2. CORE TECHNOLOGY STACK

Inference Engine: Ollama

Serves as the local execution environment for open-weight models. It allows seamless swapping of models depending on the user's hardware capabilities, ensuring strict zero-telemetry operations.

Core LLM: Llama 3 (8B)

The primary open-weight model used for zero-shot legal text analysis. Highly capable of extracting specific clauses and translating dense legal jargon into plain English summaries.

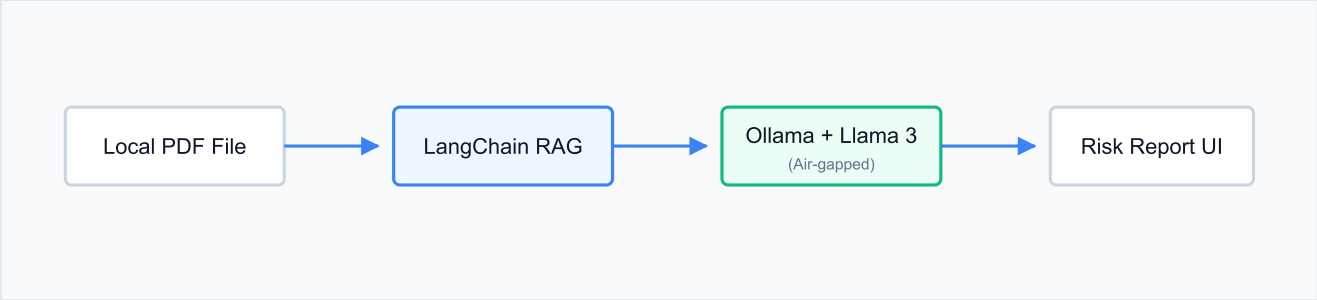
RAG Pipeline: LangChain

Orchestrates the Retrieval-Augmented Generation process. It chunks large contract PDFs, embeds them using local models, and stores them in a transient, in-memory vector database.

Agentic Framework: CrewAI

Powers the multi-agent negotiation simulator, allowing specialized local agents to debate the contract from different perspectives (Advocate vs. Corporate) to uncover hidden loopholes.

3. SYSTEM ARCHITECTURE & DATA FLOW



4. IMPLEMENTATION PHASES

Phase 1: Zero-Trust Foundation

The initial build focuses on establishing the secure, air-gapped environment. We will package a local installation script that bundles Ollama with the required system dependencies. The user interface, built in Streamlit or Gradio, will be configured to operate entirely on `localhost`. Security audits will be conducted to ensure the application actively blocks outbound network requests when a document is actively loaded in the workspace.

Phase 2: The "Red Line" Translation Engine

Once the local inference pipeline is stable, development shifts to prompt engineering and the RAG mechanism. The system will be calibrated against open-source datasets of standard legal agreements to identify deviations. We will implement specific extraction chains looking for:

- Perpetual IP transfer and "Work for Hire" clauses.
- Extended payment terms (Net-60 or Net-90).
- Overly broad non-compete definitions.
- Asymmetrical liability and indemnification requirements.

5. ADVANCED MULTI-AGENT SIMULATOR

The core innovative feature that distinguishes VaultSign AI from simple summarization tools is the Multi-Agent Negotiation Simulator. Utilizing CrewAI, the application spawns two distinct persona-driven agents that interact locally to stress-test the contract before the user ever contacts the client. This adversarial approach uncovers edge-case risks that a single-pass analysis might overlook.

Agent Role	System Prompt Objective	Value to the User
The Advocate	Identify any clause that limits the freelancer's future earning potential or delays their compensation.	Ensures the freelancer is legally protected and paid on time. Highlights immediate red flags.
The Corporate Counsel	Defend the contract as written. Justify predatory clauses as "standard industry practice."	Prepares the freelancer for the exact pushback they will receive from the client's legal team.
The Mediator	Synthesize the debate and generate a compromise clause.	Provides actionable, copy-paste ready counter-offer language that is polite but firm.

Phase 3: Pushback Generator Integration

Following the agent simulation, the system synthesizes the findings into context-aware email templates. Instead of providing raw legal advice, the tool generates polite, professional correspondence. For example, if a Net-90 payment clause is found, the generated text will read: *"To align with my standard operational cash flow, I kindly request adjusting the payment terms from Net-90 to Net-30."*

6. GO-TO-MARKET & MONETIZATION STRATEGY

TARGET AUDIENCE	DATA PRIVACY	TARGET MRR (YEAR 1)
60M+	100%	$15,000

As a designated "Side Hustle" project, VaultSign AI is designed for immediate commercial viability. By solving a high-pain problem (legal fees and NDA breaches), the application provides clear, measurable ROI for the end-user.

Monetization Tiers:

- **Basic Core (Free):** Users can download the open-source electron app and run basic RAG-based clause extraction using their own hardware and Ollama instance.
- **Pro Desktop License ($10/month):** Unlocks the CrewAI Multi-Agent Simulator, the Pushback Email Generator, and regular updates to the "Predatory Clause Database" (a curated list of new legal traps).
- **Cloud-Secure Web Version ($3/contract):** For users lacking the hardware for local inference, we will host the application on a private cloud instance utilizing open-weight models (preventing data transfer to OpenAI/Anthropic), monetized on a pay-per-use model.

> **The Open-Source Advantage:** This business model is only possible because of open-source AI. Using proprietary APIs would mean absorbing high token costs for 50-page PDFs and fundamentally breaking the user's NDA, destroying the product's primary value proposition.